KAP
ENERGY GROUP

CATALOGUE

2026

Intelligence in every watt

KAP (Kara Energy Pardazan Hooshmand) · Smart energy consumption management

SMART ENERGY. REAL IMPACT.



ABOUT KAP

Home-grown technology for smart electricity management

KAP (Kara Energy Pardazan Hooshmand) is a technology company that optimises electricity consumption through edge computing and the industrial Internet of Things, and uses smart load management to build the foundation of a virtual power plant — so that unnecessary consumption falls and the national grid stays more stable, instead of building costly new power plants.

Edge AI | IIoT | Split-Core CT | EVO-IPMVP | DSM | VPP

What we do

We measure consumption, analyse its pattern and cut wasted use with alerts and safe control.

How we measure

With non-invasive current sensors and savings verification under the international EVO-IPMVP protocol.

Why it matters

Reducing consumption replaces the heavy cost of building new power plants and new generation.



WHAT IS KAP

What is KAP?

An electricity bill only shows the result. KAP measures consumption moment by moment, identifies each device's share at different times, suggests practical ways to use less and measures the effect of every change precisely — without disrupting comfort or daily activity.



VALUE CHAIN

KAP value chain: from data to results

Measure MEASURE | **Analyse** ANALYZE | **Act** ACT | **Verify** VERIFY | **Create value** MONETIZE

KAP's output is not just a chart: it is an action with an owner, a measurable result and an auditable report.

OUR GOALS

KAP goals

Four goals, one path: from installing the first sensor to forming a virtual power plant.

01

Establish the technical chain

Fast installation, consumption measurement and clear guidance for accurate decisions; controlling costs without compromising comfort.

02

Savings with sustainable economics

Deployment in return for an energy-saving certificate contract; lower bills benefit the customer and KAP is funded by supplying saving certificates.

03

Build capacity to scale

Starting with universities to refine the method and team skills; then schools, residential complexes and regional markets.

04

Moving toward a virtual power plant

Monitoring load at peak hours and aggregating demand-management capacity nationwide, without affecting people's comfort.



VISION

KAP vision

Integrated management of electricity use across the country: linking scattered demand flows and building the foundation of a virtual power plant without new physical assets, so the energy grid becomes more resilient.



ROADMAP

Roadmap: step by step, with clear criteria

01

Universities

Pilot, data collection and a proven delivery method

02

Education, offices and homes

Wider services and automatic control

03

Industrial production

Monitoring equipment made on production lines

04

National and regional scale

Virtual power plant and neighbouring markets

Each stage starts only after technical approval and the required funding are in place.

PRODUCTS

Three products, one chain

KAP products work as one chain: data comes from the electrical panel, AI understands it and the user sees the result.



1 · Sensor

KAP smart sensor (ESA)

Installed non-invasively at the building's electricity intake, it measures current, voltage and power in real time.

2 · AI

KAP AI monitoring and control system

Analyses the data, finds consumption patterns and anomalies, and builds recommended actions.

3 · App

KAP app and dashboard

Shows everything simply so managers and users can see consumption, decide and measure the result.





SMART CONTROL

KAP control equipment

Control is applied only after precise measurement, with the user's consent and only to permitted loads; the user can always override it.



SMART ANALOG DRIVE

Smart socket and controller

Turns an evaporative cooler's two fixed speeds into continuous, flexible speed control; stable motor speed control without complex circuitry; automatic room-temperature adjustment using AI and weather data.

Alert first, then act: the user is notified before any control action and has time to decide.

SAFETY MODULE

Sensor and protection module

Connects to the electrical panel and building management system, monitors the waveform in real time and detects distortion and overvoltage, while separating critical loads from controllable ones.



An energy management system does not replace certified protective equipment or safe electrical design. Images are conceptual.

SERVICES

KAP services

From installation to the saving certificate: a complete service that expands step by step.



01 Phase one

Consumption monitoring and analysis

Continuous monitoring of electric current in the software and dashboard; when and where consumption and costs are high becomes clear.

02 Phase one

Recommendations and follow-up

Actions tailored to each site's pattern; the site manager decides and feedback is followed up.

03 Phase one

Installation, deployment and training

Panel and communications safety check, sensor installation and operator training; monitoring and evaluation then continue.

04 Phase one

Savings measurement and certificate

Results measured against the baseline, recorded in a documented file, and verified savings issued as a certificate.

05 Expansion phase

Smart control and sockets

Automatic commands only with user consent and on permitted loads, backed by reliable service and support.

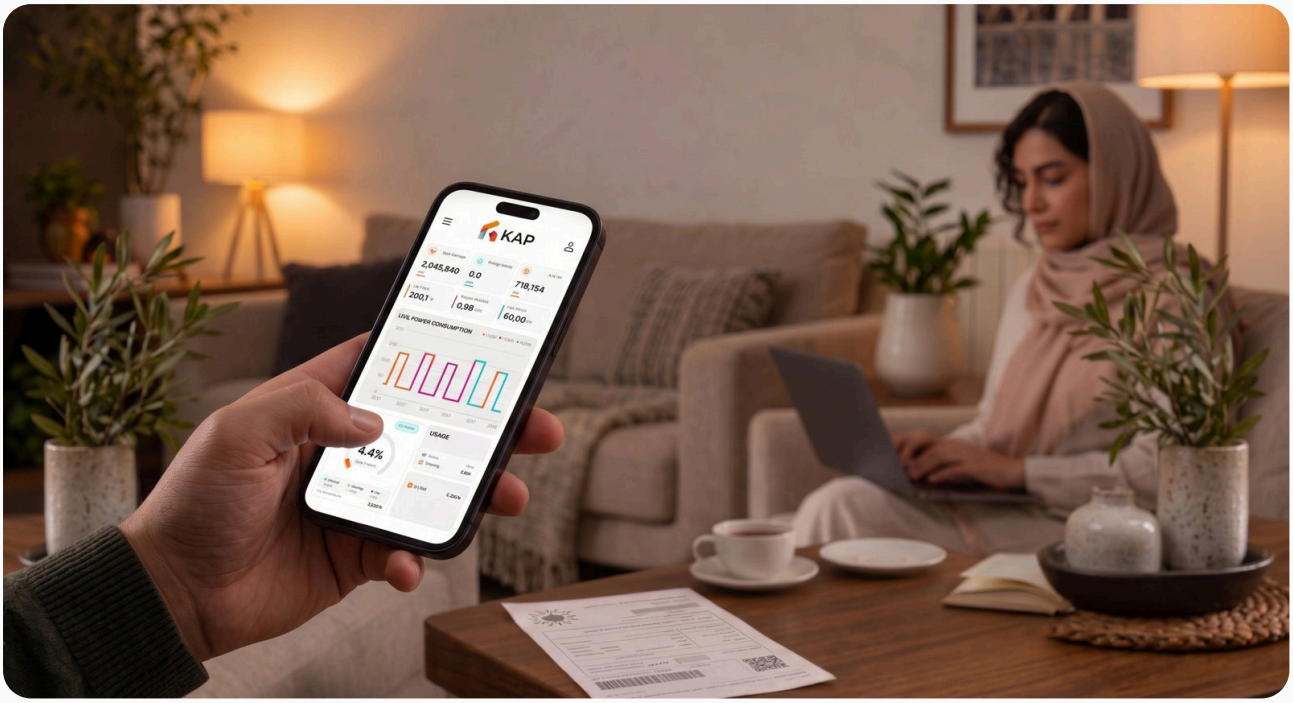
06 Expansion phase

Aggregated capacity: a virtual power plant

Coordinating demand-management capacity across many sites to reduce peak load and stabilise the grid.

HOW KAP HELPS

The electricity bill tells you the result, but not the reason.



Understand the consumption pattern

Which area uses electricity, when and for what; unnecessary operation, such as equipment left on after hours, is found.

Simple, coordinated action

Solutions suited to each space: the academic calendar on campus, residents' comfort at home and process constraints in industry. The site manager makes the final decision.

Real measurement of results

The effect of each action is measured accounting for temperature and the activity calendar, separating real reductions from shifted peak times.

WHERE KAP WORKS

Every building, an opportunity

Consumption is scheduled around each site's operating hours and actual needs.







Homes
Lower bills and alerts for unusual use

Offices and retail
Lighting and cooling control outside working hours

Universities and schools
Night and holiday consumption under watch

Industry
Peak-load management and power quality

Agriculture
Irrigation pumps, greenhouses and cold storage

Solar and generators
Smart upgrade and periodic monitoring

Power sector
Lower simultaneous load at peak hours

EV charging
Charging load coordinated with the building

THE FUTURE

The future of KAP

Three horizons, each brought into operation after technical validation and an economic case.



Five-year goal

National scale and a virtual power plant

Expanding from universities to schools and households, and coordinating load-shedding capacity nationwide to lower grid peaks.



Under study

New energy applications

Greenhouse consumption monitoring, EV charging load management and thermal-load control; each after technical and economic testing.



Programme goal

Domestic manufacturing and regional presence

Producing monitoring equipment on industrial lines and entering neighbouring markets, backed by quality certification.

KAP business model

From smart consumption to energy services

01 Smart electricity monitoring and management



Measure | Analyse | Recommend | User approval | Savings

The CT current sensor, smart processing unit and KAP app identify freed-up capacity and the economic value of savings.

Revenue: energy-saving certificates
after the certificate is approved, issued and traded

02 Smart equipment manufacturing





KAP smart socket, controller and sensor: the tools that carry out approved actions on site.

Revenue: equipment and service sales

03 Energy services development

Under study



Greenhouse thermal storage

Night-time power → heat storage → greenhouse heating · system sales and heating services



EV charging station

Development target: charging in about 5 minutes, depending on station power and battery capability · charging service revenue

Conceptual view of the business model. Energy services require technical review and the relevant permits.

KAP Sand Reactor

Night-time power, greenhouse heat

A sand thermal store converts off-peak grid electricity into heat, keeps it in a sand bed and delivers controlled heat to the greenhouse when needed.





1 m	6 m	7.5 t
inner bed diameter	effective tank length	approx. sand mass

Thermal charging

An electric heater transfers heat through the core and exchanger network into the sand bed.

Discharge after shutdown

A closed air or water loop turns the sand's energy into deliverable heat.

Greenhouse climate control

Sensors, fans and pumps regulate heat output and use low-value off-peak power.

Theoretical thermal capacity per module

754 kWh

$E = m \times c \times \Delta T \div 3600 \cdot 500 \rightarrow 100°C$

Heating a 200 m² greenhouse in the design scenario

about 17 hours

approximate discharge time, depending on climate

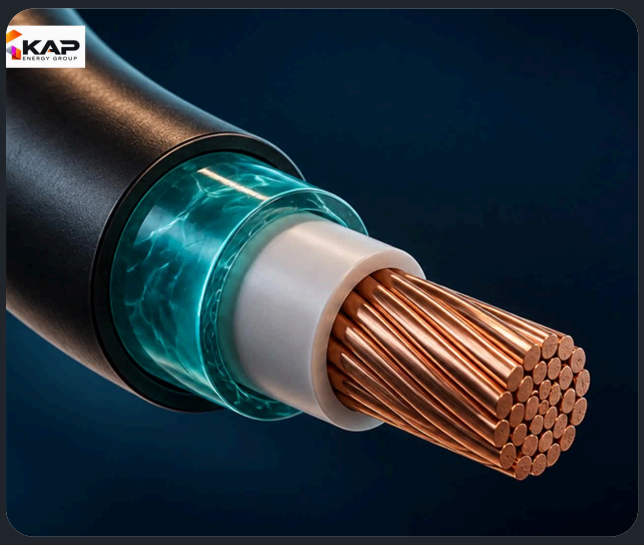
Figures are based on design calculations and a sample scenario, and will be finalised after a prototype is built and field-tested.

ULTRA-FAST CHARGING

From energy management to ultra-fast EV charging

KAP's self-cooling cable manages heat around the copper conductor, allowing higher current without a larger diameter than the reference cable.





Target transfer power

1200 kW

Target charge from 10 to 80 percent

under 5 minutes

depending on station power, temperature and battery capability

On-site power supply

Grid connection, transformer and switchboard sized for the station's simultaneous load

Storage where needed

Reducing instantaneous grid load with a station battery

Load management with KAP

Coordinating charging with the building load and logging delivered energy

Under study; final specifications will be set after joint testing with vehicle and battery makers and station operators.

TRUST

Safety, privacy and transparent measurement

Alert first, then act

No control action happens without the user's knowledge and consent.

Critical loads excluded

Essential and sensitive equipment is exempt from automatic switching.

Secure data

Encrypted transfer and storage, with three access levels: owner, technical team and system administrator.

No cameras

Everything is analysed from electricity consumption data alone.

MEASUREMENT & VERIFICATION

Savings measured under EVO-IPMVP

A measurement boundary is defined, a baseline recorded and results reported with adjustment for temperature and the activity calendar. Power (kW) and energy (kWh) are measured and reported separately. KAP does not promise the same savings percentage for every site, because results depend on each site's conditions.



WORK WITH US

Partnership model

Cost depends on the type of site. Scope and contract details are set for each site after a technical visit.

In return for saving certificates

Homes and buildings eligible for saving certificates

- Deployment in return for an energy-saving certificate contract
- Contract basis: verified savings against the baseline
- Control equipment such as smart sockets and controllers is supplied separately

Installation and periodic monitoring

Sites with solar panels or generators

- Smart upgrade of generation and consumption equipment
- Installation and commissioning fee
- Recurring monitoring and support fee

NEXT STEPS

Next step: working with KAP

01 — Visit
Understanding the site and collecting consumption data

02 — Recommend
Technical and commercial proposal tailored to the site

03 — Pilot
Trial run and result measurement

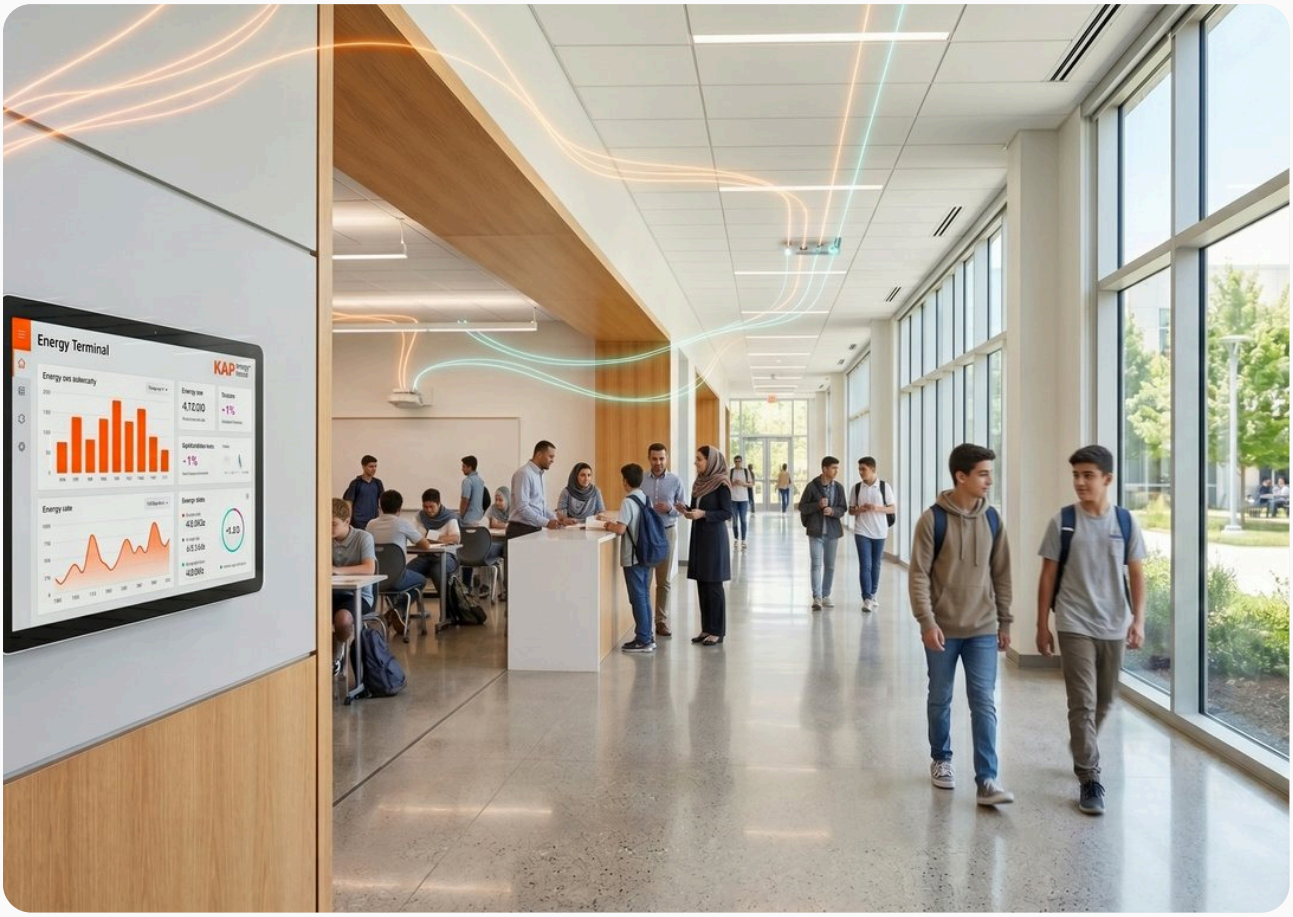
04 — Scale-up
Expansion decided on measured results



PILOT

We started at the University of Tehran

The KAP pilot has been carried out at the Faculty of World Studies, University of Tehran, and we are now looking for other faculties and universities. With their varied equipment and engaged operators, faculties are the best setting to learn and refine the method.



Pilot stages

01
Site visit and circuit mapping

02
Installation and baseline recording

03
Gradual, safe control

04
Results report

Each pilot delivers: a consumption report, a repeatable installation method, operating instructions and a realistic estimate of service cost.

OUR TEAM

Fifteen people, complementary expertise

Hardware and sensors, AI, control and automation, renewable energy, digital transformation, social sciences and international development — together for one goal: smart electricity use.



Dr. Vali Shokri



Dr. Ali-Asghar Javadi-Nezam



Prof. Ehsan Shahghasemi



Dr. Shahu Sabbar



Dr. Ali Javadi-Nezam



Mohammad-Javad Aliakbari



Fatemeh Shirkhodaei



Seyedeh Fatemeh Hashemi Kasvaei



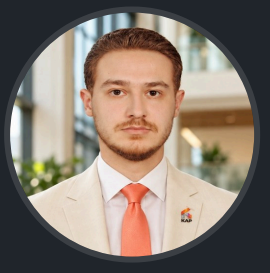
Moein Farhadi Jahroudi



Mahan Fallahpour



Mahmoud-Reza Fallahpour



Yasin Raoufi



Reza Safaei Dariani



Ensieh Farkhondehpay



Negin Ghaffari

OUR TEAM

KAP team



Dr. Vali Shokri

Education: PhD in Management · MSc in Marketing · Postdoc in Entrepreneurship Management, University of Tehran

Specialist in consumer behaviour and technology adoption; helps consumption-reduction measures change usage patterns without dissatisfying operators.



Dr. Ali-Asghar Javadi-Nezam

Education: PhD in Mechatronics and MSc in Biotechnology, Delft University of Technology (TU Delft) · BSc in Pure Mathematics, Sharif University of Technology

Experienced technology leader in international companies; designs KAP's current and power measurement systems and rugged industrial hardware.



Prof. Ehsan Shahghasemi

Education: Full Professor of Communication Sciences, University of Tehran · Postdoc, San Francisco State University (SFSU)

Specialist in the social effects of big data and AI; guides KAP's cultural and social strategy for reducing consumption at community scale.



Dr. Shahu Sabbar

Education: PhD in International Studies and Communication, Sogang University (Seoul) · Assistant Professor, Faculty of World Studies, University of Tehran

Specialist in network analysis and technology exchange in East Asia; maps KAP's path into regional markets.



Dr. Ali Javadi-Nezam

Education: Doctor of Veterinary Medicine (DVM), Islamic Azad University · Executive MBA, University of Tehran

Experienced manager in industrialisation and precision hardware manufacturing; supports quality control, product life cycle and production of KAP monitoring equipment.

OUR TEAM

KAP team



Mohammad-Javad Aliakbari

Education: Doctor of Business Administration (DBA), University of Tehran · MBA · BSc in Mechanical Engineering, Islamic Azad University (Science and Research)

Leads the business model, project governance and inter-agency coordination; turning energy-monitoring data into tradable value.



Fatemeh Shirkhodaei

Education: PhD in Environmental Engineering · BSc in Civil Engineering (Dams and Networks)

Specialist in critical infrastructure and risk management; responsible for documentation, organisational training and the strategic literature of the virtual power plant.



Seyedeh Fatemeh Hashemi Kasvaei

Education: BSc in Architectural Engineering · Executive MBA, Technology Leadership and Organisational Development

Designer of interactive smart systems; builds KAP's lasting relationships with key players in industry and the energy market.



Moein Farhadi Jahroudi

Education: BSc in Electrical Engineering (Digital Systems), College of Engineering, University of Tehran

Specialist in hardware and AI accelerators; on-panel load disaggregation (Edge NILM) and the system's cyber-physical security.



Mahan Fallahpour

Education: BSc in Computer Engineering, College of Engineering, University of Tehran

Designs deep-learning models; KAP's EVO-IPMVP-compliant baseline engine and predictive model training.

OUR TEAM

KAP team



Mahmoud-Reza Fallahpour

Education: Executive Business Administration programme, University of Tehran

Over 25 years managing industrial and energy projects; connects KAP with energy-intensive industries and renewable power plants.



Yasin Raoufi

Education: MSc and BSc in Electrical Engineering (Control), University of Tabriz

Specialist in industrial control and communications; secure relay control and demand-response support.



Reza Safaei Dariani

Education: BSc in Civil Engineering Technology

Architect of smart automation and AI agents; secure, scalable deployment of agentic systems in infrastructure.



Ensieh Farkhondehpay

Education: Doctor of Business Administration (DBA), University of Tehran · MSc in E-Commerce, Luleå University of Technology (Sweden)

Over 17 years in IT and enterprise systems management; foreign trade director and digital marketing specialist. Backbone of the KAP platform's stable, secure software pillar.



Negin Ghaffari

Education: Graduate in Architecture, Sapienza University of Rome (Italy)

Experience in cross-border projects and work with international brands; precise delivery of monitoring systems in large buildings.

LET'S TALK

Let's make energy use smart.

Leading in foresight and innovation

SMART ENERGY. REAL IMPACT.

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KAP digital card

KAP ENERGY GROUP